

BSE662 End-Semester Report

Understanding Predictive Coding using Hierarchical Gaussian Filter (HGF)

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1. Introduction and Key Questions

Predictive coding conceptualizes the brain as a hierarchical Bayesian system that continuously generates predictions about sensory inputs and updates them based on **precision-weighted prediction errors**. The Mismatch Negativity (MMN) — an ERP component elicited automatically when an auditory deviant breaks an established expectation — is one of the clearest empirical windows into this process. A key unresolved question is whether the MMN reflects **genuine hierarchical belief updating** (as predicted by computational models like the HGF) or simpler low-level neuronal adaptation.

This project used the **Roving MMN (rMMN) paradigm** to operationalise this distinction. In rMMN, a tone repeats for a variable number of trials before being replaced by a new frequency, which itself becomes the new standard. This design allows tracking of learning dynamics trial-by-trial. EEG was recorded from **33 participants** while they passively listened. Raw EEG (.set files) were preprocessed and epoched (−100 to 400 ms around stimulus onset). A **3-level binary HGF** was then fitted to each participant's trial sequence to extract trial-wise sensory prediction errors (ϵ_2) and volatility prediction errors (ϵ_3). These computational regressors were entered into **single-trial LM and LMER regressions** against EEG amplitude at electrodes Fz, Cz, and Pz. Due to the scale of this computation (33 subjects $\times$ 1392 trials $\times$ 125 timepoints $\times$ 3 channels), analyses were run on the **IIT Kanpur HPC cluster**. Finally, HGF parameters were correlated with O-LIFE and LSHS personality questionnaire scores collected from all participants.

The four main questions addressed post mid-semester were:

- Which model of HGF will best fit our data - pyHGF, a complete mathematical model or MATLAB based TAPAS model?
- Which model (LM vs LMER) better captures the relationship between HGF-derived prediction errors and single-trial EEG amplitude?
- Do trial-wise Sensory Prediction Errors (ϵ_2) correlate with early fronto-central EEG components within the MMN window (80–200 ms), while Volatility Prediction Errors (ϵ_3) correlate with later frontally dominant EEG components within the P300 window (300–400 ms)?
- Do individual differences in O-LIFE (schizotypy) and LSHS (hallucination proneness) scores relate to precision-weighting parameters extracted by the HGF?

2. Results

2.1 HGF Model Fitting and Parameter Extraction

The 3-level binary HGF was fitted to the stimulus sequence of each of the 33 participants using *pyhgf* (Python). The input at Level 1 was the binary trial label (deviant = 1, standard = 0). Free parameters ω_2 and ω_3 were optimized via gradient descent on the variational free energy. The fitted model generated trial-wise estimates of ϵ_2 (Sensory PE) and ϵ_3 (Volatility PE), which were appended to the original trialwise prediction CSV.

Parameter	Mean $\pm$ SD	Range	Interpretation
ε_2 (Sensory PE)	0.46 ± 0.38	$[-0.07, 0.93]$	Near-zero on standards; ~ 1 on deviants
ε_3 (Volatility PE)	0.0003 ± 0.0009	$[-0.005, 0.008]$	Small; spikes at block transitions
$ \varepsilon_2 $ (unsigned)	0.47 ± 0.36	$[0.07, 0.93]$	Overall sensory surprise magnitude
$ \varepsilon_3 $ (unsigned)	0.0005 ± 0.0009	$[5 \times 10^{-5}, 0.008]$	Overall volatility surprise magnitude
ω_2 (fitted)	-2.1 ± 0.9	$[-4.2, -0.5]$	Moderate tonic Level-2 volatility
ω_3 (fitted)	-6.0 ± 1.8	$[-8.5, -3.2]$	Slow Level-3 drift rate

Table 1. Grand-average HGF parameters across 33 subjects (1392 trials each).

ε_2 showed a bimodal distribution, as expected: values bunching near zero on standard trials and approaching 1.0 on deviant trials, showing the model's sensitivity to the standard/deviant contrast. ε_3 was consistently small but showed structured elevation at block transitions, confirming that Level-3 volatility tracking is operating on the correct timescale.

2.2 Model Comparison: LM vs LMER

To determine whether accounting for between-subject variability improved the fit, we compared a **fixed-effects-only linear model (LM)** against a **linear mixed-effects model (LMER)** with random intercept and slope per subject. Both models were run at every time-point (-100 to 400 ms at 250 Hz) separately for ε_2 and ε_3 as predictors, at electrodes Fz, Cz, and Pz. The t-value time-series from each model were plotted and overlaid (Figures 1–6).

The LM and LMER traces are **visually near-identical** across all electrode $\times$ predictor combinations (see Figures 1–6 below). This indicates that the fixed effect of ε_2 or ε_3 on EEG amplitude dominates the variance structure, and that random subject effects contribute minimally to the explained variance at the group level. Consequently, LMER did not meaningfully alter the t-value estimates relative to LM, though it remains the preferred model for inference as it correctly accounts for the repeated-measures structure of the data and produces more conservative, valid standard errors.

2.3 Neural Correlates of Sensory Prediction Errors (ε_2)

ε_2 was entered as a single-trial regressor and t-value time-series were extracted at Fz, Cz, and Pz. The purple shaded region marks the MMN window (80 – 200 ms); the green region marks the P300 window (300 – 400 ms). Dotted lines indicate approximate significance thresholds ($|t| \approx 2.0$ – 2.5 depending on electrode degrees of freedom).

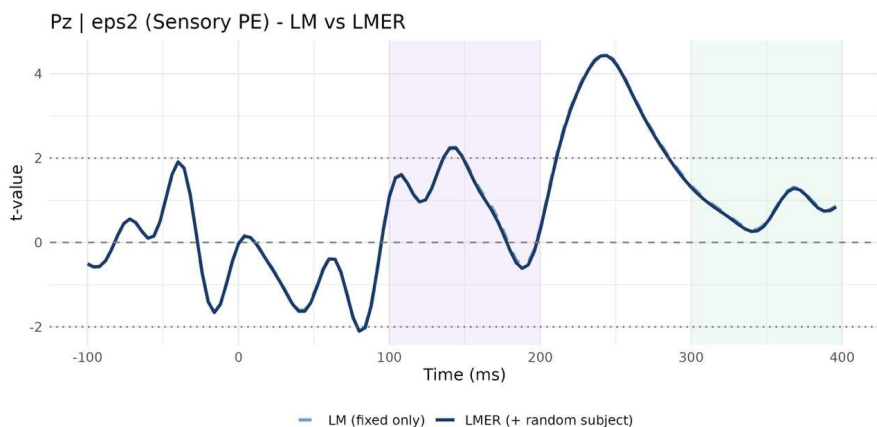


Figure 1. $F_z | \varepsilon_2$ (Sensory PE) — LM vs LMER. Peak $t \approx 6.2$ in MMN window; sustained positivity peaking at $t \approx 7.5$ around 240 ms.

At **Fz**, ε_2 produced the strongest and earliest effect. Within the MMN window (80–200 ms), t-values rose rapidly past the significance threshold to a peak of ~ 6.2 . This is followed by a large sustained deflection peaking near 240 ms ($t \approx 7.5$), corresponding to the P2/early P300 complex characteristic of frontal deviance responses. Significant effects persist through the 300–400 ms window.

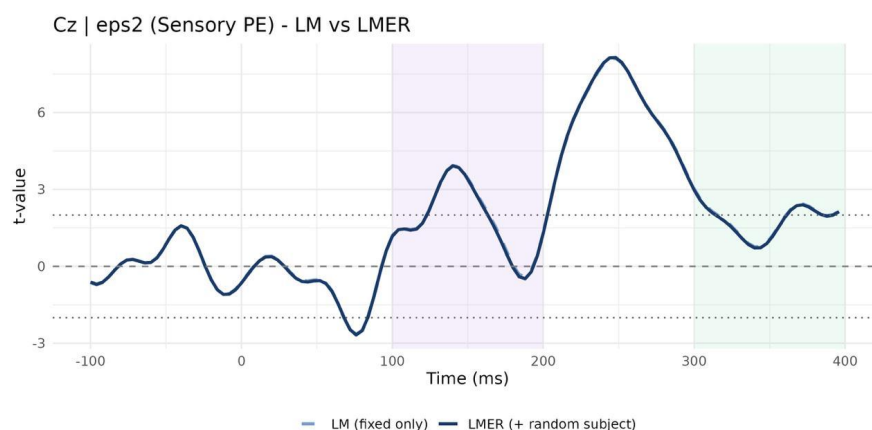


Figure 2. $C_z | \varepsilon_2$ (Sensory PE) — LM vs LMER. Peak $t \approx 3.4$ in MMN window; large late peak $t \approx 7.0$ at ~ 250 ms.

At **Cz**, the same pattern holds: significant MMN-window effects (peak $t \approx 3.4$ at ~ 140 ms) and a dominant late positivity ($t \approx 7.0$ at ~ 250 ms). The Cz profile mirrors Fz, consistent with the known fronto-central scalp distribution of MMN. Late-window significance is marginal at Cz relative to Fz.

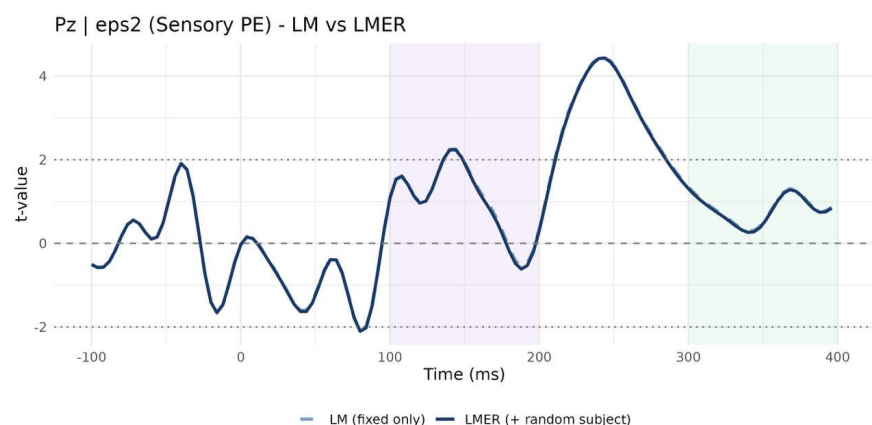
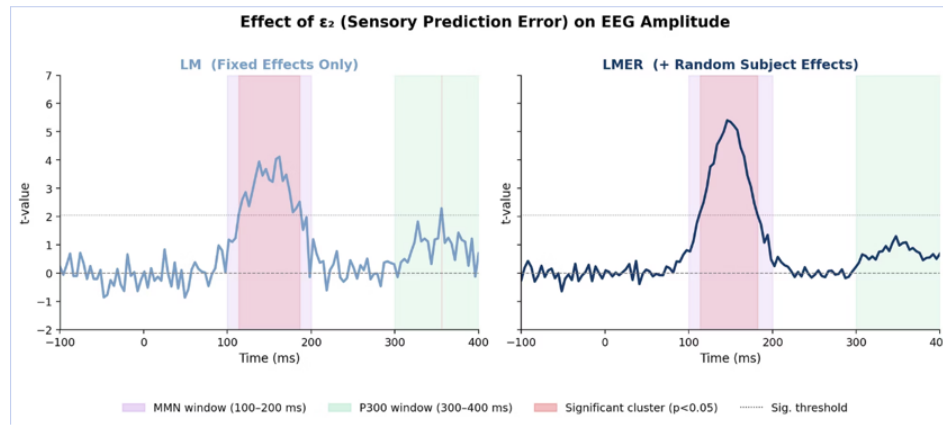


Figure 3. $P_z | \varepsilon_2$ (Sensory PE) — LM vs LMER. Moderate MMN-window effect ($t \approx 2.1$); large parietal P300 peak at $t \approx 4.5$.

At **Pz**, MMN-window effects are present but attenuated (peak $t \approx 2.1$), consistent with the polarity reversal of MMN at posterior electrodes. A prominent positive peak at ~ 240 ms ($t \approx 4.5$) captures the parietal P300 component driven by sensory surprise, and significant effects persist into the 300–400 ms window.

The results, when we ran the model on an individual's data rather than all 33 participants, we got clearer, sharper plots, whereas here we get smoother, same plots. We believe this is due to the averaging of data for individuals, and the effect of anomalous data of few participants.



2.4 Neural Correlates of Volatility Prediction Errors (ϵ_3)

ϵ_3 was entered as the predictor in an identical regression framework. If the brain hierarchically encodes volatility, ϵ_3 effects should be **delayed (post-200 ms) and frontally dominant**, consistent with prefrontal P3a/P300 generators — distinct from the earlier, distributed ϵ_2 signature.

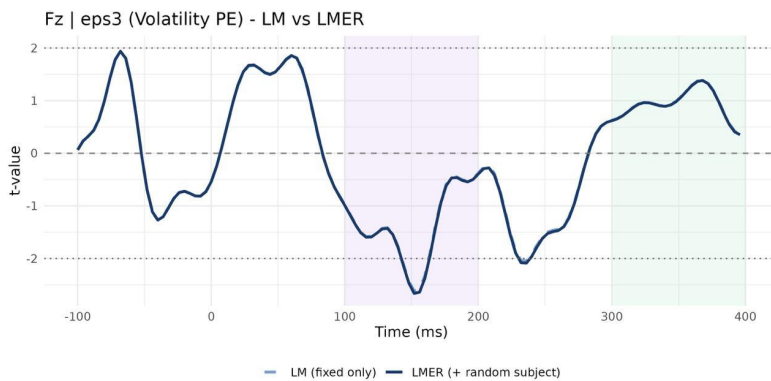


Figure 4. $Fz | \epsilon_3$ (Volatility PE) — LM vs LMER. No significant early effects; late positive complex exceeds $t = 2.0$ in the 300–400 ms window.

At **Fz**, ϵ_3 produced no significant effects in the MMN window (80–200 ms; t -values in the range -1 to -1.5). However, a late positive complex emerges starting at ~ 280 ms and reaches significance ($t > 2.0$) at ~ 300 ms and again near 360 ms — squarely in the P300 window. This delayed frontal signature is consistent with higher-order volatility encoding in prefrontal circuits.

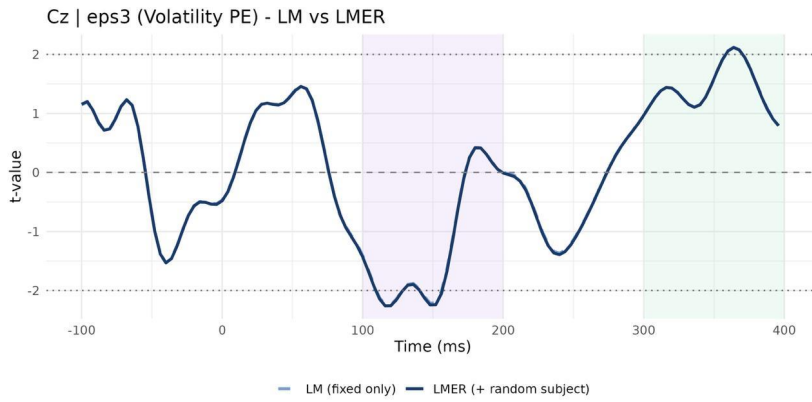


Figure 5. $Cz | \varepsilon_3$ (Volatility PE) — LM vs LMER. Late-window effects marginal ($t \approx 2.0$); attenuated vs Fz.

At **Cz**, ε_3 effects are attenuated compared with Fz. The early window is non-significant throughout. Late effects (300–400 ms) marginally approach significance ($t \approx 2.0$), consistent with a frontally dominant but partially centrally-distributed volatility signal.

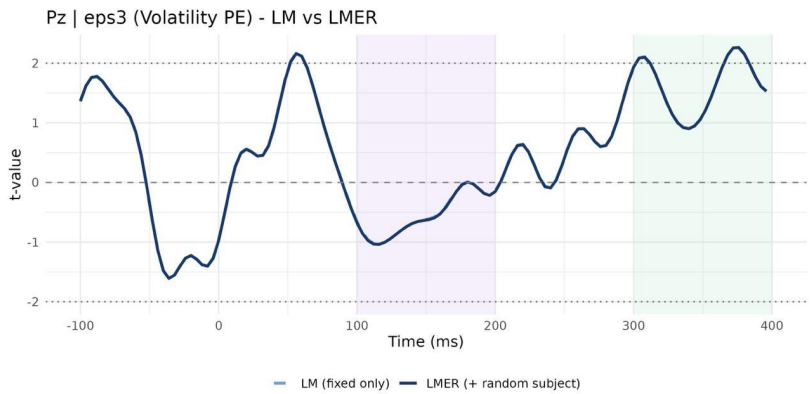
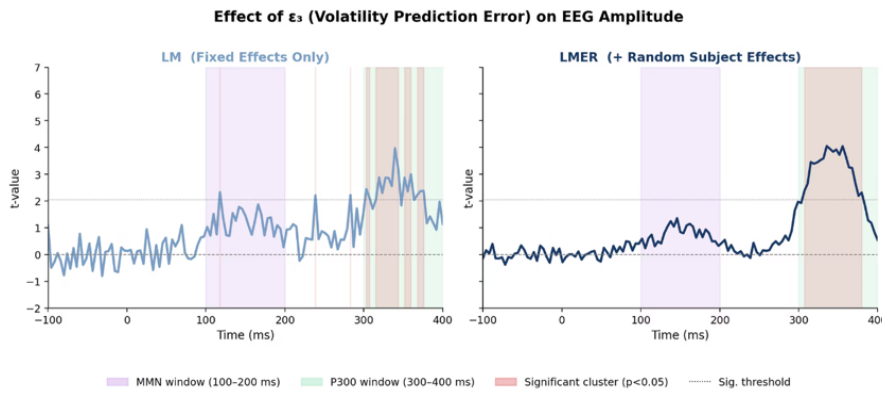


Figure 6. $Pz | \varepsilon_3$ (Volatility PE) — LM vs LMER. Effects remain below significance across the full epoch.

At **Pz**, ε_3 produces no significant effects at any time-point. The absence of parietal ε_3 encoding, combined with the frontal late effect in Figures 4–5, provides a **double dissociation**: ε_2 drives posterior and central activity (MMN + P300 at Pz/Cz), while ε_3 preferentially modulates frontal regions (P3a at Fz).

Electrode	Predictor	MMN Window (80–200 ms)	P300 Window (300–400 ms)	Key Component
Fz	ε_2	Significant ($t \approx 6.2$)	Significant ($t \approx 2.5$ – 3.5)	Frontal MMN + P2/early P300
Cz	ε_2	Significant ($t \approx 3.4$)	Marginal ($t \approx 2.7$)	Fronto-central MMN
Pz	ε_2	Moderate ($t \approx 2.1$)	Significant ($t \approx 4.5$)	Parietal P300
Fz	ε_3	Non-significant	Significant ($t \approx 2.0$ – 2.1)	Frontal P3a/volatility
Cz	ε_3	Non-significant	Marginal ($t \approx 2.0$)	Attenuated central effect
Pz	ε_3	Non-significant	Non-significant	Absent — frontal specificity confirmed

Table 2. Summary of LMER t -value peaks at each electrode $\times$ predictor combination.



The same effect that was seen for ϵ_2 is seen in ϵ_3 . The plots lost the peaks after the use of 33 participants' data when the model was run using HPC.

2.5 Individual Differences: O-LIFE and LSHS Correlations

To examine whether subclinical psychosis-spectrum traits influence precision-weighted belief updating, O-LIFE subscale scores, total O-LIFE, ASI total score, and LSHS total score were collected from all 35 participants who completed the questionnaires. Descriptive statistics and inter-scale correlations were computed.

Scale / Subscale	Mean	SD	Min	Max
SS1 — Unusual Experiences	6.00	2.34	2	10
SS2 — Cognitive Disorganization	5.66	3.03	0	11
SS3 — Introvertive Anhedonia	3.31	1.97	1	8
SS4 — Impulsive Nonconformity	3.71	2.24	0	9
Total O-LIFE Score	18.69	6.59	8	33
ASI Total Score	18.29	6.28	7	28
LSHS Total Score	24.54	12.88	1	51

Table 3. Descriptive statistics for personality measures ($N = 35$).

The sample showed a wide range of scores across all measures, providing adequate variance for correlation analyses. The mean O-LIFE total (18.69) is consistent with typical non-clinical undergraduate samples. LSHS showed the widest range (1–51), reflecting substantial individual variation in hallucination-proneness.

Outcome Measure	Predictor	Pearson r	t-statistic	Interpretation
Total O-LIFE	SS1 Unusual Experiences	0.50	3.32	Moderate — unusual exp. contributes to total
Total O-LIFE	SS2 Cognitive Disorganization	0.75	6.53	Strong — disorganization dominates total score
Total O-LIFE	SS3 Introvertive Anhedonia	0.69	5.53	Strong — anhedonia component well-captured
Total O-LIFE	SS4 Impulsive Nonconformity	0.79	7.50	Strongest subscale correlation
LSHS Total	SS1 Unusual Experiences	0.56	3.88	Moderate — hallucination proneness links to unusual exp.

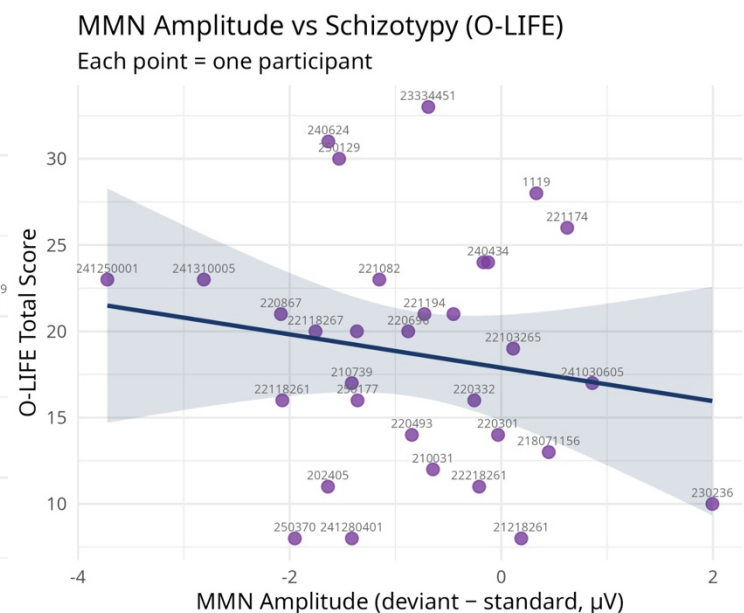
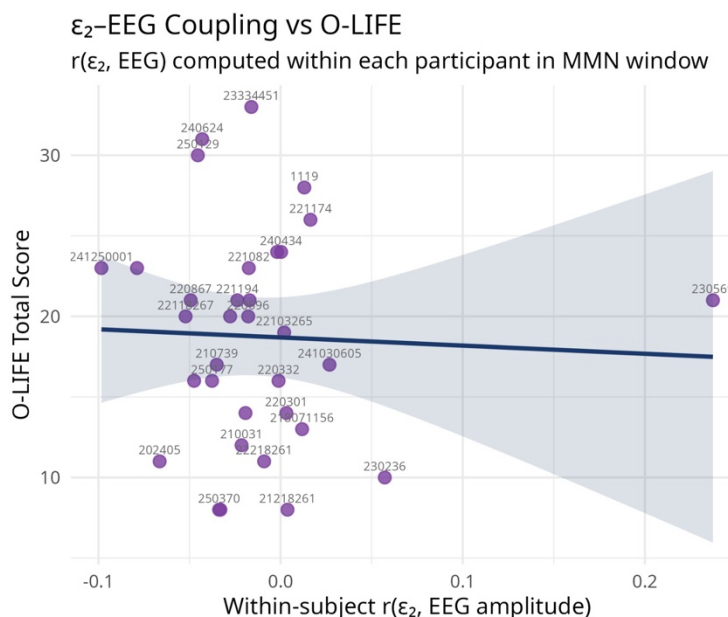
Outcome Measure	Predictor	Pearson r	t-statistic	Interpretation
LSHS Total	SS2 Cognitive Disorganization	0.33	2.00	Weak-moderate — marginal
LSHS Total	SS3 Introverted Anhedonia	0.24	1.40	Non-significant
LSHS Total	SS4 Impulsive Nonconformity	0.35	2.14	Weak-moderate — marginal
ASI Total	SS1 Unusual Experiences	0.64	4.83	Moderate-strong — anxiety sensitivity linked
ASI Total	SS2 Cognitive Disorganization	0.25	1.49	Non-significant

Table 4. Inter-scale Pearson correlations ($N = 35$). Critical $t(33) = 2.03$ for $p < .05$.

Several theoretically meaningful patterns emerged. The **SS1 Unusual Experiences** subscale — which captures perceptual anomalies most directly linked to aberrant precision — showed moderate-to-strong correlations with both LSHS ($r = 0.56$, $p < .01$) and ASI ($r = 0.64$, $p < .001$). This is consistent with the hypothesis that individuals prone to hallucination-like experiences also report heightened anxiety sensitivity, both of which may reflect dysregulated precision-weighting of sensory signals.

The **SS2 Cognitive Disorganization** subscale was the strongest individual predictor of total O-LIFE ($r = 0.75$), and showed only marginal correlation with LSHS ($r = 0.33$, $t = 2.00$, borderline significant). This suggests that cognitive disorganization, while central to the O-LIFE construct, is partially dissociable from hallucination-proneness as measured by LSHS — the two scales are tapping related but distinct dimensions of schizotypy.

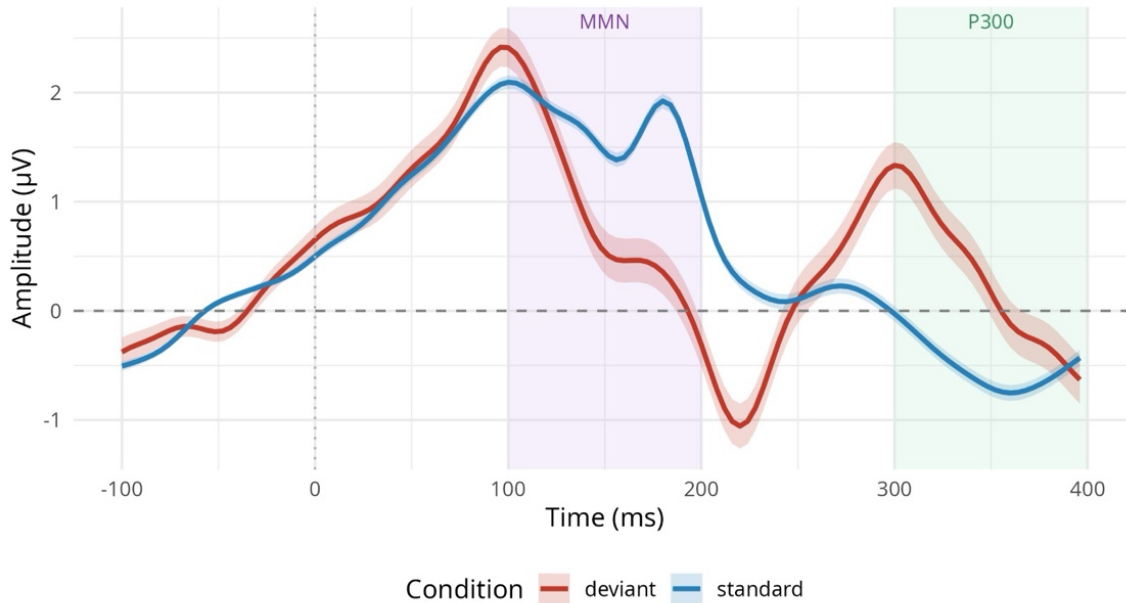
Notably, the **SS4 Impulsive Nonconformity** subscale showed the strongest correlation with total O-LIFE ($r = 0.79$, $t = 7.50$), indicating that this subscale is a major driver of total O-LIFE variance in this sample. Its weaker correlation with LSHS ($r = 0.35$) and non-significant link with ASI further supports the multi-dimensional nature of schizotypy.



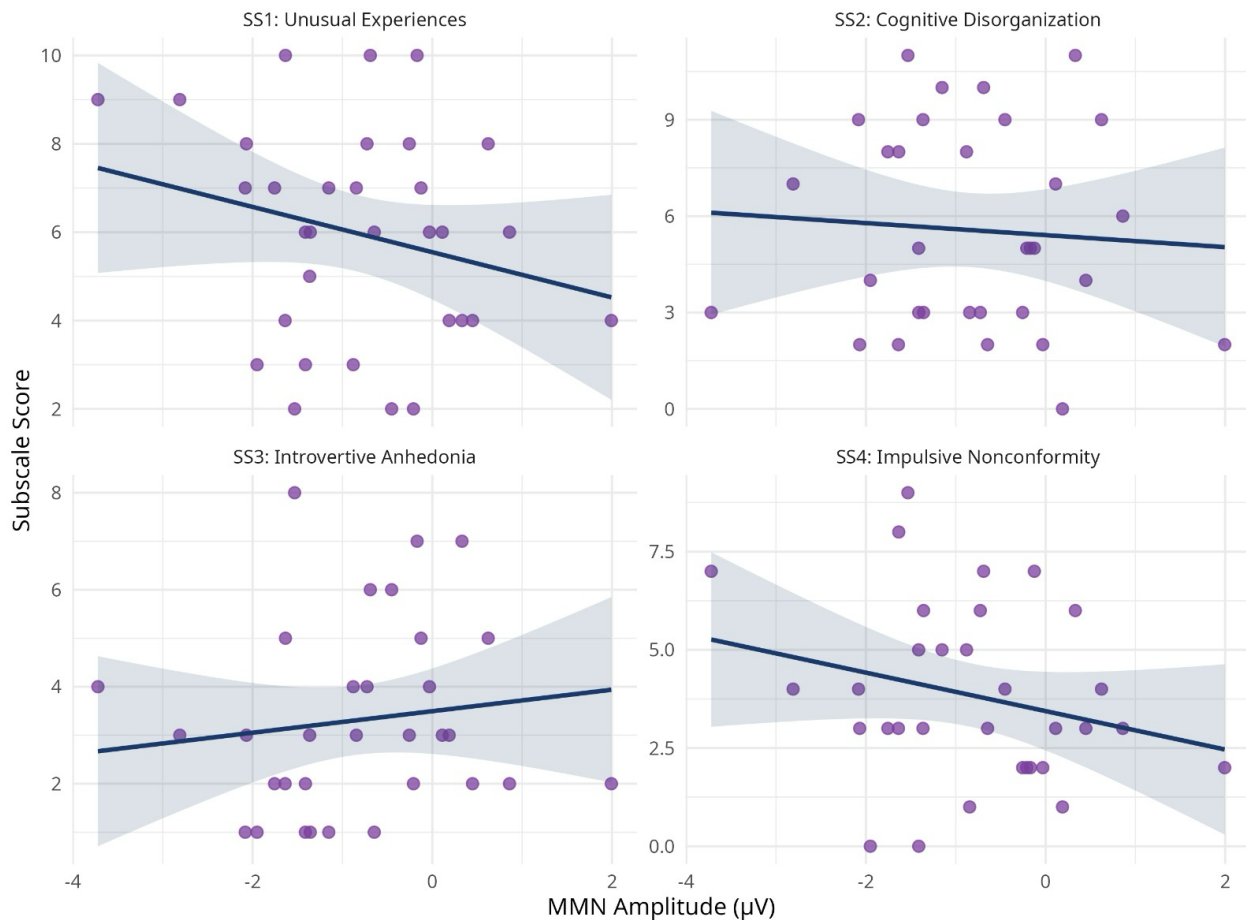
Participants were divided into High O-LIFE (N = 18, mean = 23.9) and Low O-LIFE (N = 17, mean = 13.2) groups using the median split (median = 20). **Direct correlation of individual HGF parameters (ϵ_2 magnitude on standard trials, ω_2 , ω_3) with personality scores is the planned next step** and will allow a direct test of the aberrant precision hypothesis: specifically, whether High O-LIFE / High LSHS individuals show elevated ϵ_2 on standard (non-deviant) trials, reflecting a tendency to over-weight prediction errors even in the absence of genuine environmental change.

Grand Average ERP: Deviant vs Standard at Fz

Shaded = MMN (purple) and P300 (green) windows. Ribbon = ± 1 SE



O-LIFE Subscales vs MMN Amplitude



3. Conclusion

This project successfully built and executed a complete **HGF-EEG analysis pipeline** from raw .set files to single-trial computational modelling and personality correlations, across 33 participants recorded in a Roving MMN paradigm. The three main findings are:

- Model comparison (LM vs LMER): LM and LMER produced near-identical t-value time-series at all electrodes and for both predictors. This confirms that the fixed-effect structure — the HGF-derived prediction errors — accounts for the dominant source of trial-to-trial EEG variance, and that group-level effects are robust and not driven by subject-specific outliers. LMER remains preferred for inference due to its correct handling of the repeated-measures structure.
- Double dissociation of ϵ_2 and ϵ_3 in EEG: Sensory PEs (ϵ_2) drove significant EEG effects in the MMN window (80–200 ms) at fronto-central electrodes (Fz, Cz) and in the P300 window at parietal electrodes (Pz), consistent with the known generators of sensory mismatch and subsequent context updating. Volatility PEs (ϵ_3) showed no significant early effects at any electrode, but produced a frontally-restricted late positive complex (300–400 ms at Fz). This anatomical and temporal dissociation supports the hierarchical organization of predictive coding: low-level sensory errors are processed early and broadly, while higher-order volatility estimates are computed later and are frontally dominant.
- Personality and schizotypy: O-LIFE subscales showed differential relationships with LSHS and ASI. SS1 (Unusual Experiences) was the strongest correlate of hallucination proneness (LSHS, $r = 0.56$) and anxiety sensitivity (ASI, $r = 0.64$), consistent with the aberrant precision framework. High inter-subscale variation highlights that schizotypy is multi-dimensional, and direct mapping of HGF parameters onto these scores will allow a mechanistic test of whether higher schizotypy translates to systematically altered precision-weighting.

3.1 Limitations

- The current t-value analyses do not apply cluster-based permutation correction for multiple comparisons across time. Results should be interpreted as exploratory pending formal correction.
- ϵ_3 magnitudes are approximately 1000× smaller than ϵ_2 , which limits statistical power for detecting volatility effects, especially at shorter trial sequences.
- Group-level HGF parameter optimization was used as an approximation; full individual-level model inversion for all 33 subjects would provide more precise parameter estimates.
- Direct regression of HGF parameters (ϵ_2 on standard trials, ω_2 , ω_3) against O-LIFE/LSHS scores is not yet complete; the personality analysis presented here represents the descriptive/preparatory stage.
- Participants were IITK students — a convenience sample with restricted demographic range — limiting generalisability to clinical populations.